



Valentin ALLARD

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Research interests

My research focuses on the properties and dynamics of **nuclear superfluids in neutron stars**, with the goal of improving nuclear models for extrapolation to **unexplored regions of dense matter**. I study systems ranging from **ultracold quantum gases** which share close similarities with neutron condensates in the crust of neutron stars, to the neutron–proton **superfluid mixture** in the core. I use time-dependent **nuclear energy density functional theory** to provide **self-consistent microscopic inputs** for macroscopic dynamical models of neutron stars. I aim to understand how these superfluid phases influence astrophysical phenomena such as **pulsar glitches**, **neutron star cooling**, and **gravitational wave emission**.

Keywords: Nuclear astrophysics, Neutron stars, Superfluidity, Superfluid hydrodynamics, Ultracold quantum gases, Density Functional Theory, Many-body physics, Low-energy nuclear reactions, Soft X-ray transients, Neutron star cooling.

Academic background

- **2024—2026: Assistant professor**

Warsaw University of Technology (Faculty of Physics) – Warsaw (Poland)

Research project: *Pairing dynamics in nuclear collisions*

Principal Investigator: Piotr Magierski

Study of the nuclear pairing on superheavy element production through the study of collisions between $^{48}\text{Ca}+^{250}\text{Cf}$ and $^{124}\text{Sn}+^{178}\text{Yb}$. Simulations are performed using the [LISE](#) package which numerically solves the 3D time-dependent Hartree-Fock Bogoliubov on a spatial lattice and without symmetry restrictions. The package is run on [LUMI](#) and [TSUBAME](#) supercomputers.

- **2020–2024: PhD candidate**

Université Libre de Bruxelles (ULB) – Brussels (Belgium)

Dissertation: [*Superfluid dynamics in neutron stars*](#)

Supervisor: Nicolas Chamel

Jury: Michel Tytgat, Wouter Ryssens, Anthea Francesca Fantina, Piotr Magierski

Grant: 4-year grant from F. R. S. – FNRS, Research project (PdR T.004320)

Graduation: May 21, 2024

PhD thesis mainly focused on effects of temperature and currents in hot neutron-proton superfluid mixtures studied within the framework of nuclear energy-density functional theory. The impact of superfluid dynamics on the crust cooling of transiently accreting neutron stars is also investigated.

- **2019–2020: Teaching qualification for upper secondary education - Physics**

Université Libre de Bruxelles (ULB) – Brussels (Belgium)

Magna cum laude (average grade 15.25/20)

- **2017–2019: Master's degree in Physics**

Université Libre de Bruxelles (ULB) – Brussels (Belgium)

Master thesis : *Entrainment effects in neutron stars cores* (mark : 17.50/20)

Advisor: Nicolas Chamel

Summa cum laude (average grade 16.96/20)

- **2014–2017: Bachelor's degree in Physics**

Université de Namur – Namur (Belgium)

Magna cum laude

Publications

Statistics (source: Google Scholar, September 2025)

- **Citations:** 95
- **h-index:** 7
- **i10-index:** 5

Publications in peer-reviewed international journals

- *Evidence of Gapless Superfluidity in MXB 1659-29 With and Without Late Time Cooling*, V. Allard and N. Chamel. Published in [Universe 2025, 11\(5\), 140 \(2025\)](#)
- *Gapless neutron superfluidity in the crust of the accreting neutron stars KS 1731-260 and MXB 1659-29*, V. Allard and N. Chamel. Published in [Eur. Phys. J. A 60 116 \(2024\)](#)
- *Gapless Neutron Superfluidity Can Explain the Late Time Cooling of Transiently Accreting Neutron Stars*, V. Allard and N. Chamel. Published in [Phys. Rev. Lett. 132, 181001 \(2024\)](#)
- *Gapless superfluidity in neutron stars – Normal fluid fraction*, V. Allard and N. Chamel. Published in [Phys. Rev. C 108, 045801 \(2023\)](#)
- *Gapless superfluidity in neutron stars - Thermal properties*, V. Allard and N. Chamel. Published in [Phys. Rev. C 108 015801 \(2023\)](#)
- *1S_0 Pairing Gaps, Chemical Potentials and Entrainment Matrix in Superfluid Neutron-Star Cores for the Brussels–Montreal Functionals*, V. Allard and N. Chamel. Published in [Universe 7, 470 \(2022\)](#)
- *Entrainment effects in neutron-proton mixtures within the nuclear energy-density functional theory. II. Finite temperatures and arbitrary currents*, V. Allard and N. Chamel. Published in [Phys. Rev. C 103, 025804 \(2021\)](#)
- *Entrainment effects in neutron-proton mixtures within the nuclear energy-density functional theory: Low-temperature limit*, N. Chamel and V. Allard. Published in [Phys. Rev. C 100, 065801 \(2019\)](#)

Conference Proceedings

- *Gapless superfluidity and Neutron Star cooling*, V. Allard and N. Chamel. Published in [Phys. Sci. Forum 2023, 7\(1\)](#)

Scientific Conferences & Research visits

Oral contributions to international conferences

- **The Modern Physics of Compact Stars and Relativistic Gravity 2025**
23–26 September 2025 (Yerevan State University, Armenia)
Talk: [Gapless superfluidity and neutron stars](#)

- **ECT*, Nonequilibrium phenomena in superfluid systems: atomic nuclei, liquid helium, ultracold gases, and neutron stars**
12–16 May 2025 (*ECT*, Trento, Italy*)
Talk: [Superfluid hydrodynamics in neutron stars](#)
- **EOS Manaslu Workshop**
3–4 July 2023 (*KU Leuven, Belgium*)
Talk: [Gapless superfluidity in neutron stars](#)
- **Nonequilibrium Phenomena in Superfluid Systems**
27–29 March 2023 (*Warsaw University of Technology, Poland*)
Talk: Gapless superfluidity in neutron stars
- **Conference on Quantum-Many-Body Correlations in Memory of Peter Schuck (QMBC 2023)**
21–23 March 2023 (*IJCLab, Orsay, France*)
Talk: [Superfluid dynamics in neutron stars](#)
- **56th Karpacz Winter School**
24–28 February 2020 (*Karpacz, Poland*)
Talk: [Entrainment effects in neutron-proton superfluid mixtures within the nuclear energy-density functional theory](#)

Poster contributions to international conferences

- **XMM-NEWTON 2024 Workshop - “The X-Ray Mysteries Of Neutron Stars And White Dwarfs”**
5–7 June 2024 (*ESAC, Madrid, Spain*)
Poster: [Evidence of gapless neutron superfluidity from the late time cooling of transiently accreting neutron stars](#)
- **Doctoral School Joliot-Curie – “Nuclear Matter Under Pressure”**
4–9 September 2022 (*Saint-Pierre d’Oléron, France*)
Poster: [Gapless superfluidity and neutron star cooling](#)
- **PHAROS Conference 2022 – “The multi-messenger physics and astrophysics of neutron stars”**
16–19 May 2022 (*Sapienza – University of Rome, Italy*)
Poster: $^1\text{S}_0$ pairing gap and entrainment for BSk functionals in neutron star cores
- **Royal Astronomical Society Early Career Poster Exhibition**

14–28 September 2020 (*online*)

Poster: [Entrainment effects in neutron-proton mixtures in neutron star cores](#)

Research visits

- Visit to Nuclear Physics Division (Piotr Magierski, Daniel Pęczak & Gabriel Wlazłowski)
18 January–8 February 2023 (*Warsaw University of Technology, Poland*)
Focus: Dynamics of a nucleus in a neutron superfluid.
- Visit to Nuclear Theory Group (Kazuyuki Sekizawa)
28 July 2025 (*Nuclear Theory Group — Science Tokyo, Japan*)
Focus: New perspectives in the study of neutron stars

Teaching & Student Supervision

Teaching

- **2019–2023:** tutorial classes (in French) for first-year bachelor students, course [PHYS-F-103 – Physique](#), Université Libre de Bruxelles, 5 ECTS, 24h per academic year.
- **2021–2023:** tutorial classes (in French) for second-year bachelor students, course [PHYS-F-205 – Physique 2](#), Université Libre de Bruxelles, 5 ECTS, 24h per academic year.

Supervision of Master students

- **2022–2023, S. Brausch** (Université Libre de Bruxelles): *Study of collective modes in a superfluid neutron star.*